INCH-POUND

MIL-PRF-83818A (USAF)
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SUPERSEDING
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15 October 1990

PERFORMANCE SPECIFICATION

TRUCK, FORKLIFT, 4,000 AND 6,200 POUND CAPACITY, FRONT/SIDE LOADING, INTERNAL COMBUSTION ENGINE

This specification is approved for use by the Department of the Air Force and is available for use by all the Departments and Agencies of the Department of Defense.

1. SCOPE

1.1. Scope. This document covers a front/side loading, sit-down operator, internal combustion engine powered forklift.

1.2 Classification. Forklifts should be one of the following types, as specified (see 6.2b):

Type I: 4,000 pound capacity at 24 inch load center
Type II: 6,200 pound capacity at 24 inch load center

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in section 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the applicable issue of these documents are those listed in the specific issue of the

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use improving this document should be addressed to: WR-ALC/LVR, 225 Ocmulgee Court, Robins AFB GA 31098-1647, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2.c).

SPECIFICATIONS

FEDERAL

A-A-52557 - Fuel Oil, Diesel; For Posts, Camps and Stations

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoD Index of Specifications and Standards (DoDISS) cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2.c).

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B56.1- Safety Standard for Low Lift and High Lift Trucks

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017)

ELECTRONIC INDUSTRIES ASSOCIATION (EIA)

IS-647 - Requirements for the Control of Electromagnetic Interference Emissions and Susceptibility Characteristics of Equipment Intended to Operate in Severe Electromagnetic Environments

IS-648 - Measurement of Electromagnetic Interference Characteristics of Equipment Intended to Operate in Severe Electromagnetic Environments

(Application for copies should be addressed to Electronic Industries Association, Engineering Department, 2500 Wilson Boulevard, Arlington, VA 22201.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE J98 - Personnel Protection for General Purpose Industrial Machines
SAE J185 - Construction and Industrial Equipment, Access Systems for
SAE J386 - Operator Restraint System for Off-Road Work Machines
SAE J833 - Human Physical Dimensions
SAE J925 - Minimum Service Dimensions for Off-Road Machines

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SAE J994 - Alarm, Backup, Electric Performance Test and Application
SAE J1511 - Steering for Off-road, Rubber-Tired Machines

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096)

THE TIRE AND RIM ASSOCIATION, INC. (TRA)

The Tire and Rim Association Yearbook

(Application for copies should be addressed to The Tire and Rim Association, Inc., 175 Montrose West Avenue, Suite 150, Copley, OH 44321)

3. REQUIREMENTS.

3.1 First article. When specified (see 6.2.d), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Materials.

3.2.1 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.2.2 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other.

3.3 Operating environment. The engine shall start within 5 minutes at any ambient temperature between -25° F and +125° F without any form of external power or heat source. The truck shall operate as specified herein within 15 minutes after engine start.

3.4 Winterization. When specified (see 6.2.e), the truck shall be furnished with type A or type C winterization. Measures to achieve operation at the temperatures specified for each type of winterization may include but shall not be limited to coolant heater, engine oil heater, heavy duty battery, battery heater, thermal insulation in cab, glow plugs, measured shot ether injection, air dryer (air brake system only), and arctic type lubricants. If the use of engine coolant, engine oil or battery heaters is elected for winterization, then all heaters shall operate on 110 volt, alternating current. A three wire, 25 foot long cable of adequate capacity for all heaters being used simultaneously shall be provided. If heaters are used, a means to control output level shall be provided to ensure that the temperature of engine coolant and/or engine oil does not exceed +150° F, and battery electrolyte temperature does not exceed +80° F.

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3.4.1 Type A winterization. Type A winterization shall allow the truck to start and operate in an ambient temperature of -65° F, and maintain the operator's cab at an internal temperature of at least +40° F at cab floor level.

3.4.2 Type C winterization. Type C winterization shall allow the truck to start and operate in an ambient temperature of -25° F, and maintain the operator's cab at an internal temperature of at least +40° F at cab floor level.

3.5 Human factors. The truck shall be operable and maintainable by personnel ranging from a small person normally clothed through a large person arctic clothed in accordance with SAE J833 and SAE J925. This shall include all travel, load motion, and steering controls, entry to the operator's seat, and access to all fluid level check points. To accommodate operation by small persons, the force on the service brake necessary to meet the requirements of 3.37.10 shall not exceed 80 pounds. The force required to fully set the parking brake (see 3.37.11) shall not exceed 70 pounds if hand operated or 150 pounds if foot activated. With no load on the truck, engine at idle speed, and truck not moving, the tangential force on the steering wheel required to cause the truck steering wheels to turn shall not exceed 10 pounds.

3.6 Safety. The truck shall conform to the applicable requirements of ASME B56.1, SAE J98, and OSHA standards in effect at the time of issue of the solicitation.

3.6.1 Seat belts and operator's restraint. An operator restraint system conforming to SAE J386 shall be installed.

3.6.2 Backup alarm. When specified (see 6.2.f), a backup alarm conforming to SAE J994, type A, B or C shall be furnished.

3.7 Rated load capacity. The truck shall be designed for safe operation with the rated capacity load at all lift heights, fork tilt and fork positions that the truck is capable of achieving. The rated load capacity for the Type I shall be 4,000 pounds. The rated load capacity for the Type II shall be 6,200 pounds. For rating purposes, the rated load shall be considered to be a four-foot cube with the center of gravity at the geometric center of the cube. Because of certain ordnance loads encountered in operational use, the truck shall be designed and manufactured to meet specific stringent structural load-bearing safety margins. A structural safety factor (based on yield strength) of at least 3.0 times the rated load, or the maximum dynamic load multiplied by 2, whichever is greater, shall be met (see 6.4).

3.8 Engine. The truck shall be powered by a standard commercial diesel engine, or if otherwise specified (see 6.2.g), the engine shall be a gasoline or liquid petroleum gas (LPG), or dual fuel (gasoline and LPG) type. The diesel engine shall operate on diesel fuel conforming to A-A-52557 and JP-8 (see 6.6) without detrimental effect on the engine, or its performance.

3.8.1 Engine exhaust. The exhaust system shall incorporate a muffler, and shall terminate at a location clear of the operator's station. The outlet shall not be directed toward the ground and a means to prevent rain water intrusion into the exhaust system shall be furnished.

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3.9 Fuel system. The truck shall be capable of continuous operation for a minimum of eight hours without refueling. The fuel tank filler neck opening shall be capable of interfacing with a fuel filler tube with an outside diameter of 2.90 inches. The fuel tank shall be capable of being fully drained to allow the truck to be air transportable.

3.10 Transmission. The transmission shall provide at least one forward and one reverse speed. The transmission shall provide for positive inching or declutch control throughout the entire engine rpm range, in both forward and reverse directions. The inching or declutch control shall permit lifting of rated load, at maximum engine speed, while the transmission is in a forward or reverse gear, with no vehicle motion.

3.11 Electrical system.

3.11.1 Starter switch. The starter switch shall not activate the engine starter while the engine is running, nor at any time when the engine is not running and the transmission is in any forward or reverse gear.

3.11.2 Lighting. The truck shall have sufficient exterior illumination to be operated at night in both forward and rearward modes. Instrument panel lighting shall be provided.

3.11.3 Batteries. Except when Type A winterization is specified (see 6.2.e), the batteries shall be the maintenance-free type.

3.11.4 Master switch. A keyless master switch, with a corrosion resistant identification plate, shall be installed in a convenient location making it possible to disconnect the battery from all electrical loads during the performance of maintenance actions.

3.11 5 Wiring schematic. A corrosion resistant plate containing the electrical schematic of the forklift truck shall be installed in an area where normal usage will cause it no damage. The plate may be located behind a panel that can be opened by hand.

3.12 Hydraulic system. If the truck design incorporates a hydraulic system, the requirements of 3.12.1 and 3.12.2 shall apply.

3.12.1 Hydraulic reservoir. The hydraulic reservoir filler opening shall be capable of interfacing with a filler tube with an outside diameter of 1.94 inches.

3.12.2 Hydraulic schematic. A corrosion resistant plate containing the hydraulic schematic of the truck shall be permanently installed in an area where normal usage will cause it no damage. The plate may be located behind a panel that can be opened by hand.

3.13 Lift system. Because of certain loads encountered in operational use, the truck shall be designed and manufactured to meet specific stringent lift system safety margins and shall be equipped with a fail-safe lift system which maintains safe control of the rated load if electrical,

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hydraulic or pneumatic system failure occurs. To prevent damage to military weapons during lifting and lowering operations, drift in the lift system components shall be limited to a maximum drift rate of 0.5 inch per hour (see 6.4).

3.14 Fork assembly. To successfully interface with military pallets and shipping containers, the fork assembly shall have the following dimensions and restrictions:

3.14.1 Fork spacing. The fork spacing on both the Type I and Type II trucks shall be adjusted laterally by manually moving the forks. When the forks are positioned as close together as possible, the distance between the outer edges of the forks shall be a maximum of 13.0 inches. When both forks are moved as far apart as possible, the distance between the outer edges of the forks shall be a minimum of 37.0 inches.

3.14.2 Fork size.

(All measurements in inches.)	Type I	Type II
a. Length	42.0 + or - 1.0	42.0 + or - 1.0
b. Maximum width	5.06	6.07
c. Maximum thickness	1.82	2.07
d. Maximum thickness at tip	0.5	0.5
e. Minimum bottom taper, from tip rearward	14.0	14.0

3.15 Load backrest. A load backrest shall be provided, either as an integral part of the fork carriage, or as a removable part thereof. It shall be approximately as wide as the fork carriage. The backrest shall be capable of supporting loads as high as 42.0 inches, but shall not extend more than 46.0 inches above the top surface of the forks. The upper 17 inches of the load backrest on the Type II truck shall be removable and replaceable without the need for tools. It shall be secured with quick release pins when installed. Two devices (one on each side) to restrain the load shall be located on the load backrest 16 to 20 inches above the load carrying surfaces of the forks. The devices shall have a inside diameter large enough to accept a chain and hook of 2.0 inches diameter. The thickness of the devices shall allow a hook with an inside dimension of 0.63 inch to fasten to it. The devices shall have a structural safety factor of at least 3 to 1 when a forward horizontal load, equal to the rated load, is equally applied to both devices.

3.16 Pivot and lateral motion assembly. The pivot and lateral motion of the mast or forks shall be provided to allow at least 90 degrees pivoting of load, and full lateral motion. Pivot and lateral motion shall be controlled by the seated operator. With the forks pivoted 90 degrees, the lateral motion shall be sufficient to allow the load backrest to be positioned at least 1.0 inch beyond the side of the truck. With the forks pivoted 90 degrees and moved laterally as necessary, the truck

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shall be capable of carrying 16 inch wide by 24 inches high by 20 feet long loads within the side plan outline of the truck.

3.17 Special tools. The contractor shall supply any special tools or equipment peculiar to, and only used on this forklift. One set of tools is required for each truck.

3.18 Load handling controls. All load motion controls shall be controlled by the operator's right hand and shall be self-centering, such that they return to the neutral position when released.

3.19 Cab. The truck shall be equipped with an enclosed cab that shall incorporate an overhead guard as specified in ASME B56.1. There shall be a minimum clearance of 39 inches from seat to underside of guard, when measured in accordance with ASME B56.1. The cab shall incorporate emergency exit requirements conforming to SAE J185. An hourmeter and tachometer shall be included in the instrument panel.

3.20 Workmanship. The truck shall be constructed in accordance with commonly accepted industrial workmanship standards.

3.21 Parking brake. A parking brake independent from the service brakes shall be provided.

3.21.1 Parking brake indicator. An indicator light that illuminates when the parking brake is applied shall be provided.

3.22 Horn. A horn or similar means of giving emergency warning to pedestrians or drivers of other vehicles shall be provided.

3.23 Steering. If power assisted steering is necessary to comply with the requirements of 3.5, it shall meet emergency steering requirements as specified in SAE J1511.

3.24 Towing hitch. A pintle-type towing hitch shall be installed at the center rear of the truck. The center of the hitch opening shall be positioned to accommodate loads with their towing ring approximately two feet above the ground. The pintle shall be capable of sustaining a rearward, horizontal force of at least 3 times the gross vehicle weight (GVW) of the truck.

3.25 Towing lugs. Towing lugs of sufficient diameter to accept common chains and hooks of 3.0 inches diameter shall be furnished so that the truck can be towed from the front. The lugs shall be capable of sustaining a forward, horizontal force of at least three times the GVW of the truck.

3.26 Tires. Tires and rims shall conform to the Tire and Rim Association recommendations. Tires selected shall permit the truck to travel over improved and unimproved surfaces in weather conditions including ice and snow.

3.27 Noise limits. The sound level measured at the operator's station shall not exceed 94 dB(A). If the sound level is between 84 and 94 dB(A), see 6.2.h.

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3.28 Preparation, primer and paint. All exterior surfaces and all interior surfaces normally painted shall be prepared and primed with procedures and materials compatible with the final paint. Unless otherwise specified (see 6.2i), the final paint shall be a standard, commercially available, polyurethane, color commonly known as "forest green".

3.28.1 Desert sand. When specified (see 6.2i), the final paint shall be a standard, commercially available, polyurethane, no-gloss tan color.

3.28.2 Gray. When specified (see 6.2i), the final paint shall be a standard, commercially available, polyurethane, light gray color.

3.28.3 Color of markings. Markings for green trucks shall be black paint, and desert sand trucks shall be white paint. Lusterless gray trucks shall have no markings. The following markings shall be applied in the indicated locations:

a. Vehicle capacity ("4,000 POUNDS" for Type I, or "6,200 POUNDS" for Type II) in 1.5 inch high letters on the lifting arms, mast, or front of truck.

b. "NO RIDERS" on each side of truck in 2 inch high letters.

3.29 Item identification. A corrosion resistant plate shall be permanently installed on the instrument panel, or another visible, protected location. The following information shall be on the plate:

Truck, Lift, Fork
Capacity: "4,000 pounds at 24 inch load center" for Type I, and "6,200 pounds at 24 inch load center" for Type II
National stock number:
Contract number:
Serial number:
Model number:
Registration number:
Manufactured by:
  Name:
  Address:
US Property:

3.30 Instruction plates. All warning and instruction plates describing safety items or procedures to be followed during operation and servicing shall be included.

3.31 Load Chart. A corrosion resistant plate, showing the load capacity of the truck at all boom extensions and lift heights shall be permanently installed in a location clearly visible to the operator.

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3.32 Maintainability. Provisions shall be made for adjustment, servicing, or replacement of parts and components using common tools. Dimensions of hand or arm access openings shall be in accordance with SAE J925 for arctic clothing.

3.33 Servicing. All lubrication points and fluid reservoirs shall be easily accessible for inspection and servicing and shall be located in a manner such that servicing personnel exposure to injuries and burns is minimized.

3.34 Transportability. The truck shall be capable of being transported by water, rail and highway. It shall also be air transportable in C-130, C-141, C-5 and C-17 aircraft without removal of any components (see 6.5).

3.35 Weights and dimension. Limiting weights and dimensions shall be as follows:

a. Maximum width: 45 inches for Type I and 54 inches for Type II.

b. Maximum height, with mast a lowest position: 88 inches for Type I and 91 inches for Type II. This includes mast and overhead guard, whichever is tallest.

c. Maximum weight: 13,000 pounds on each axle.

3.36 Slinging and tie downs. Permanently installed slinging and tiedown devices that enable the truck to be lifted in its normal travel position and to be secured to the load surface of a transportation medium shall be provided. The attachment points shall be designed to interface with aircraft attachment points through the use of MB-2 and CGU-3E cargo tie down chains (25,000 pound capacity), MB-1 and CGU-4E cargo tie down chains (10,000 pound capacity), and CGU-1B cargo straps (5,000 pound capacity) (see 6.5.1). The attachment points shall have a structural safety factor of 2.3 to 1, based on static load. The notation "LIFT HERE" or "TIE DOWN" shall be stenciled near each lifting and tie down device, in the color specified in 3.28.3. There shall be a permanently installed, corrosion resistant, metal plate that shows the center of gravity location in the air-transport mode, and tie down instructions in C-130 and C-141 aircraft.

3.37 Performance.

3.37.1 Lifting speed. The forks with rated load shall be able to raise at a speed of not less than 55 feet per minute (fpm) over the entire distance from ground level to maximum lift height. With the engine idling at engine manufacturer's idle speed, the unloaded forks shall be able to raise at a speed of not less than 5 fpm over the entire lift range.

3.37.2 Lowering speed. The lowering speed of the forks with rated load shall be not more than 80 fpm. The lowering speed of the unloaded forks shall be not less than 40 fpm.

3.37.3 Travel speed. The travel speed with rated load shall be at least 8 miles per hour (mph) in forward and reverse direction.

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3.37.4 Lift height. The lift height, with rated load on the forks and the forks horizontal, shall be not less than 152 inches when measured from ground to top surfaces of level forks.

3.37.5 Slope ascension. The truck, with and without rated load, shall be capable of ascending a 15 percent grade at a minimum of 2 mph when the load is in the uphill direction. It shall also be capable of accelerating up-slope from a complete stop to at least 2 mph on a 15 percent grade.

3.37.6 Drift. The lift assembly shall be capable of holding the rated load at maximum lift height for 60 minutes with not more than 0.5 inch of vertical drift, with the truck stabilized at ambient temperature.

3.37.7 Right angle turn. The truck, when carrying rated load, shall be capable of backing through a turn in either direction between two parallel walls no further apart than 175 inches for Type I and 200 inches for Type II.

3.37.8 Fork tilt. With no load on the forks, the minimum rearward tilt shall be 6.0 degrees, and the minimum forward tilt shall be 3.0 degrees.

3.37.9 Ground clearance. The truck, with and without rated load on the forks, shall have sufficient ground clearance to drive from one horizontal surface, up a 17 percent ramp, then onto a second horizontal surface, with only the tires contacting any surface.

3.37.10 Service brakes. The service brakes shall stop the truck with rated load, within the distance specified in the Stopping Distance Method of ANSI B56.1, with not more than 80 pounds of force on the brake pedal (see 3.5). The service brakes shall meet the Drawbar Drag Method of ANSI B56.1, except a drawbar drag equal to 25 percent of truck gross weight shall be produced with not more than 80 pounds of force on the brake pedal (see 3.5).

3.37.11 Parking brake. The parking brake shall be capable of holding the fully loaded truck, on a 15 percent grade, in both up and down slope directions. The force required to fully set the parking brake shall be not more than 70 pounds, if hand actuated, or 150 pounds if foot activated (see 3.5).

3.37.12 Electromagnetic interference (EMI). The truck shall meet the EMI requirements (emissions and susceptibility) as specified in EIA IS-647.

3.37.13 Fork side shifting. The forks shall be capable of side shifting a total of at least 12.0 inches.

3.37.14 Fork visibility. The operator, without leaving his seat and while maintaining full control of the truck, shall be able to see at least one empty fork tip when the forks are on the ground and in the straight ahead position.

4. VERIFICATION

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4.1 Classification of inspections. The inspections shall be classified as follows:

a. First article inspection and demonstration (see 4.2).

b. Conformance inspections (see 4.3)

4.2 First article inspection. The first article truck and its components shall be compared to Section 3 of this document to verify compliance through examination and testing in the order specified below.

4.2.1 First article examination. The first article truck shall be examined in accordance with 4.5.

4.2.2 Noise level. Measure noise level at operator's ear with the engine operating at maximum rpm and while lifting rated load. Nonconformance with 3.27 shall be cause for rejection. If noise level is between 84 and 94 dB(A), the noise warning plate required by 3.27 shall be on all trucks delivered on this contract.

4.2.3 Electromagnetic interference (EMI). Test the truck to in accordance with EIA IS-648 to determine compliance with 3.37.12. Failure to meet the requirements of 3.37.12 shall be cause for rejection.

4.2.4 First article operational tests. Test the first article truck in accordance with 4.6.

4.2.5 First article cold starting test. With all components of the truck stabilized at an ambient temperature of -25° F, the engine shall start within five minutes without assistance from any external power source. Failure to start within five minutes shall be cause for rejection.

4.2.6 First article environmental tests. Operate the truck at -25° for a minimum of twenty hours, then operate at +125° for a second twenty hours using an approved test method (see 6.2.j). The truck shall be capable of operating at least eight hours a day for a cumulative total of 40 hours without failure. In the context of this paragraph, a failure is considered to be any malfunction that cannot be corrected within 30 minutes by adjustment, repair or replacement, which may cause cessation of operation, degradation of performance capabilities, damage to the truck by continued operation, or personnel safety hazard.

4.3 Conformance inspections. Conformance inspection shall include the examination described in 4.5 and the tests described in 4.6.

4.4 Requirements cross-reference matrix. Table I provides a cross-reference matrix of the section 3 requirements tested or verified in the paragraphs below.

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TABLE I: Requirements cross-reference matrix

REQUIREMENT	VERIFICATION	REQUIREMENT	VERIFICATION
3.1	4.2	3.24	4.5
3.2	4.5	3.25	4.5
3.3	4.2.5, 4.2.6	3.26	4.5
3.4	4.5	3.27	4.2.2
3.5	4.6.10, 4.6.11, 4.6.16, 4.2.6	3.28	4.5
3.6	4.5	3.29	4.5
3.7	4.6.18	3.30	4.5
3.8	4.5	3.31	4.5
3.9	4.2.6	3.32	4.6.16
3.10	4.2.6	3.33	4.5
3.11.1	4.6.17	3.34	4.6.20
3.11.2	4.5	3.35	4.5, 4.6.20
3.11.3	4.5	3.36	4.5
3.11.4	4.5	3.37.1	4.6.3
3.11.5	4.5	3.37.2	4.6.3
3.12	4.5	3.37.3	4.6.4
3.13	4.6.6, 4.6.19	3.37.4	4.6.2
3.14.1	4.6.12	3.37.5	4.6.5
3.14.2	4.6.12	3.37.6	4.6.6
3.15	4.5, 4.6.14	3.37.7	4.6.7
3.16	4.6.15	3.37.8	4.6.8
3.17	4.5	3.37.9	4.6.5
3.18	4.5	3.37.10	4.6.10
3.19	4.5	3.37.11	4.6.11
3.20	4.5	3.37.12	4.2.3
3.21	4.5	3.37.13	4.6.13
3.22	4.5	3.37.14	4.6.9
3.23	4.5		

4.5 Examination of product. The forklift truck shall be examined for compliance with requirements for materials, winterization, safety, seat belts, backup alarm, engine, engine exhaust, hydraulic system, load backrest, fork assembly, cab, battery, lighting, load handling controls, towing hitch, towing lugs, tires, parking brake, horn, slinging and tie downs, master switch, overhead guard, special tools, wiring schematic, paint, weight, markings, identification, instruction plates, load charts, servicing, dimensions, steering, and workmanship.

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4.6 Operational tests.

4.6.1 Test conditions. Unless otherwise specified (see 6.2j), inspections shall be performed in accordance with the following test conditions:

4.6.1.1 All tests shall be conducted at the ambient temperature and climatic conditions existing at the place of demonstration.

4.6.1.2 Only that maintenance established by the contractor shall be performed during testing.

4.6.1.3 For test purposes, the rated load for the truck shall be a 4,000 pound (+200, - 0 pounds) cube (for Type I) or 6,200 pound (+200, - 0 pounds) cube (for Type II), 48 inches on each side, with a center of gravity located at the geometric center of the cube.

4.6.2 Lift mechanism. With rated load on the forks, with forks level and at their maximum lift height, measure from ground to top surface of one fork to determine maximum lift height. Nonconformance with 3.37.4 shall be cause for rejection. Raise the rated load from ground level to maximum lift height and lower back to ground. This shall be considered one cycle. Repeat until 100 cycles have been completed. This is a severe test of the lift system so raising and lowering may be stopped after every 10 cycles to allow cooling of the lift system. Inability to complete the test, binding or erratic movement, any failure of the lifting system, or overheating of the system shall be cause for rejection.

4.6.3 Lifting and lowering speed. Measure distance from ground to top surface of forks in fully lowered position. Measure time required to raise rated load from ground to maximum lift height. Subtract distance from ground to top surface of forks from maximum lift height, and calculate lifting speed. Lower rated load, at maximum lowering speed (fully open lowering control) and abruptly stop the load at 1 to 2 foot height. Record this time. Measure distance from ground to top surface of one fork. Subtract this figure from the maximum lift height and calculate loaded lowering speed. Remove load and raise forks to maximum lift height. Measure time required to lower forks to ground level. Calculate the unloaded fork lowering speed. Nonconformance with 3.37.1 or 3.37.2 shall be cause for rejection.

4.6.4 Travel speed. Operate the truck at maximum speed in both forward and reverse directions with rated load in load carry position (see 6.3.2). Inability to obtain the speed specified in 3.37.3 shall because for rejection.

4.6.5 Slope ascension. Drive the truck forward with rated load in uphill position onto a ramp, hill, or other suitable surface with a slope specified in 3.37.5. Stop the truck on the slope, then accelerate up the slope reaching a steady, measured, speed. Inability to reach the speeds specified in 3.37.5 shall be cause for rejection. With rated load, drive the truck from a horizontal surface onto a ramp as specified in 3.37.9 then onto a second horizontal surface. Repeat this procedure without a load. Any portion of the truck other than the tires touching the ground in either loaded or unloaded condition shall be cause for rejection.

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4.6.6 Drift. Raise rated load to maximum lift height, with truck at ambient temperature. Place forks in horizontal position and shut off engine. Measure from ground to top surface of one fork. Let truck sit for time specified in 3.37.6. Measure again the same distance to same point on fork. Nonconformance with 3.37.6 shall be cause for rejection.

4.6.7 Right angle turn. Start with rated load in the carry position, but with forks level, and the front of the load against a wall (or a line on the floor) and the forklift perpendicular to the wall. Back the truck through a right angle, 90 degree turn. Mark the point where the outside envelope of the truck is farthest from the wall during the turn. Measure the perpendicular distance from this point to the wall. Repeat this procedure except turning in the opposite direction while backing. Exceeding the maximum allowed dimension in 3.37.7 shall be cause for rejection.

4.6.8 Fork tilt. On a level surface, raise unloaded forks to approximately 4 feet above ground level. Tilt forks full forward and measure angle of tilt. Tilt forks to full rearward position and measure tilt angle. Nonconformance with 3.37.8 shall be cause for rejection.

4.6.9 Fork visibility. Position unloaded forks on the ground and straight ahead to verify that the visibility requirements of 3.37.14 can be met. Failure to meet 3.37.14 shall be cause for rejection.

4.6.10 Service brakes. Demonstrate that the service brakes can stop the truck with rated load, and can meet the drawbar drag test, as specified in 3.37.10. Failure to meet all requirements of 3.37.10 shall be cause for rejection.

4.6.11 Parking brake. Demonstrate that the parking brake can hold the truck, with rated load on the forks, in both forward and reverse directions, on the slope specified in 3.37.11, for at least 3 minutes. Measure the force required to apply the parking brake. Nonconformance with all requirements of 3.37.11 shall be cause for rejection.

4.6.12 Fork spacing and size. When forks are adjusted to maximum and minimum separation, the distance between the outer edges of the forks shall not exceed the distance specified in 3.14.1 Measure both forks' length, width, thickness, thickness at tip, and bottom taper. Nonconformance with 3.14.2 shall be cause for rejection

4.6.13 Fork side shift mechanism. . Side shift forks to extreme right, and then to extreme left positions, measure total travel. Nonconformance with 3.37.13 shall be cause for rejection. The forks shall be side shifted with the rated load from extreme left to extreme right position, and back to extreme left. This shall be considered one cycle. Repeat until 100 cycles have been completed. This is an extreme test of the side shift mechanism and side shifting may be stopped after 20 cycles to allow cooling of the mechanism. Any binding, erratic movements, inability to complete the test, failure of the system, or overheating of the mechanism shall be cause for rejection.

4.6.14 Load backrest restraints. Apply a forward, horizontal force of 3.0 times rated load to load restraint devices for 10 minutes. The test load may be applied to all devices at once, or 3.0 times the proportional share of the load may be applied to each device. Remove load and inspect

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restraint devices, welds and supporting structure for deformation, cracks, broken welds, etc., which shall be cause for rejection.

4.6.15 Pivot assembly. With rated load on forks, pivot forks to most extreme pivoted position. Measure to ensure forks are pivoted to the amount specified in 3.16. With forks in this position, move assembly to most outward extreme lateral position. Measure distance that load backrest is extended beyond side of truck to ensure the backrest is extended the amount specified in 3.16. Pick up a load that is 16 inches wide, by 24 inches high, by 20 feet long, and place it on a secure position on the forklift. Ensure that this load does not extend beyond the side of the truck. Nonconformance with 3.16 shall be cause for rejection.

4.6.16 Maintainability. Test the truck in accordance with SAE J925 to determine compliance with 3.32.

4.6.17 Starter disconnect switch. Start the engine and let it run for at least 10 seconds. Energize the starter switch (move to "off" then "on" position if applicable) while the engine is running. Any evidence of starter engagement while the engine is running shall be cause for rejection. With the engine not running, place transmission selector in the lowest forward gear and energize the starter switch. Repeat for all forward and reverse gears. Any evidence of starter engagement while transmission is in any gear shall be cause for rejection.

4.6.18 Overload. Unless otherwise specified in the contract, these procedures will be accomplished only during first article testing. Perform a stress analysis of the truck to determine all critical stress points (see 6.3.1). The entire stress analysis is not necessary, just the location of the critical stress points, and the cause of these stresses; 3 times static (rated) load or 2 times maximum dynamic load. Apply strain gauges or other suitable instrumentation at all these critical stress points to measure stress during this test. Place truck on level surface and position supports under the frame or front axle to relieve load on front tires. Secure the rear of the truck to compensate for the additional test load. Fully extend boom, raise forks at least 24 inches above ground level with the forks centered on the carriage. For the test load use the greater of 3 times the static load of 3,000 pounds, or 2 times the maximum dynamic load, as determined in 3.7. Apply this test load to the forks. The center of gravity of the test load shall be located 24 inches forward of the front backrest surface. Let this overload remain on the forks for at least 10 minutes. Remove the load and inspect truck structure, frame, mast or boom assembly, and hydraulic system for deformation, cracks, broken welds, and hydraulic fluid leaks. Inspect forks a minimum of 6 inches on either side of heel by magnetic particle or dye penetrant method. Record readings from strain gauges and convert to, and record, the corresponding stresses. Deformation, cracks, broken welds, or hydraulic fluid leaks shall be cause for rejection

4.6.19 Fail safe hydraulic lift system. This test shall be accomplished when the lift system is hydraulically operated. Raise the rated load to maximum lift height with forks tilted back about five degrees. Measure vertical distance from ground to top surface of one fork. No personnel shall be under or in front of the load during the next step. Disconnect hydraulic line(s) that supply pressure to the lift system. Remeasure vertical distance from ground to same point on fork surface. A drop of more than 1.0 inch shall be cause for rejection. Measure fork tilt. Disconnect

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hydraulic line(s) that supply pressure to prevent forward tilting. Remeasure fork tilt. A change of more than 2.0 degrees of tilt shall be cause for rejection. Use the over-ride provided to lower rated load to ground.

4.6.20 Air transportability verification. Verify the truck is air transportable in accordance with 3.34. Verify that the truck meets all dimension and weight limitations of 3.35. Inability to meet all of these requirements shall be cause for rejection.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2.k). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES.

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The truck should be used primarily to load and unload palletized cargo in narrow aisles. It will also be used to handle loads of various sizes and weights, up to the rated load. It will also be used to handle loads measuring 16 inches wide, by 24 inches high, by 20 feet long. The truck will operate on hard, paved surfaces.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of the specification.

b. Type of forklift required (see 1.2).

c. Issue of DoDISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.2.1 and 2.3).

d. When first article is required (see 3.1).

e. When type A or type C winterization is required (see 3.4).

f. When a back up alarm is required (see 3.6.2)

g. When the engine should be other than diesel type (see 3.8).

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h. If the sound level is between 84 and 94 dB(A), the contractor should provide a corrosion resistant hazardous noise warning plate stating that operator's hearing protection is required when operating the truck. The plate should have black lettering on a yellow background and located to be clearly visible as the operator enters the cab, or is seated at the controls (see 3.19).

i. When final paint color should be other than forest green (see 3.28).

j. Proposed test method (see 4.2.6).

k. Packaging requirements (see 5.1).

6.3 Definitions:

6.3.1 Critical Stress Point. A point, or area, where the stress is greater than 50% of the yield strength of the material, when the truck has the rated load (see 3.7 and 4.6.18) on the forks.

6.3.2 Load carry position. The normal position of forks while traveling with the rated load. To be determined by the forklift manufacturer.

6.4 Nuclear certification. Refer to Air Force Instruction 91-103, Air Force Nuclear Safety Certification Program for detailed guidance.

6.5 Air Transportability. Refer to Air Force Systems Command Design Handbook 1-11, Air Transportability for detailed guidance.

6.5.1 Tie down devices. Refer to MIL-T- 25959C, Tie Downs, Cargo, Aircraft, for interface information on the MB-1, MB-2, CGU-4E, and CGU-3E tie down chains. Refer to MIL-T-27260B, Tie Down, Cargo, Aircraft, CGU-1B, for interface information on CGU-1B tie down straps.

6.6 JP-8 fuel. JP-8 fuel (NATO Code F-34) is almost identical to ASTM D1655 Jet A-1 fuel except that JP-8 has additives such as a corrosion inhibitor/lubricity improver and a fuel system icing inhibitor additive. See MIL-T- 83133, Turbine Fuels, Aviation, Kerosene Types, NATO F-34 (JP-8) and NATO F-35 for detailed information.

6.7 Subject term (key word) listing.

diesel engine driven
material handling
pivot
swingmast

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6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodian:
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Preparing activity:
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Agent:
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INSTRUCTIONS

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3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waiver any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-PRF-83818A	2. DOCUMENT DATE (YYMMDD) 97/06/15

3. DOCUMENT TITLE
TRUCK,FORKLIFT,4000 AND 6200 POUND CAPACITY,FRONT/SIDE LOADING,INTERNAL COMBUSTION ENGINE

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON *(If applicable)*	e. DATE SUBMITTED *(YYMMDD)*

8. PREPARING ACTIVITY

a. NAME WR-ALC/LVR	b. TELEPHONE *(Include Area Code)* (1) Commercial: (912) 926-1183 (2) AUTOVON: 468-1183
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